# week4_notebook_student

October 27, 2026

## 1 HS4002 Week 4

### 1.1 Task for the day

We'll investigate three relationships: 1. Age and annual income 2. Income and cultural omnivorousness 3. Race and highest education level attained

For each question: (a) visualize, then (b) compute a bivariate statistic.

[ ]:
```
# !pip install pandas numpy scipy plotnine

import pandas as pd
import numpy as np
import plotnine as p9
import warnings
warnings.filterwarnings('ignore')
from scipy import stats
```

### 1.2 Data Preparation

[ ]:
```
raw_df = pd.read_csv("gss2022_mini.csv")

chosen_variables = [
        "id", "age", "yrlvmus", "yrartxbt", "yrmovie", "yrcreat",
        "income16", "prestg10", "degree", "race"
]
df = raw_df[chosen_variables].dropna().copy()

income_map = {
        1: 500,    2: 2000,   3: 3500,   4: 4500,   5: 5500,   6: 6500,
        7: 7500,   8: 9000,   9: 11250,  10: 13750, 11: 16250, 12: 18750,
        13: 21250, 14: 23750, 15: 27500, 16: 32500, 17: 37500, 18: 45000,
        19: 55000, 20: 67500, 21: 82500, 22: 100000, 23: 120000, 24: 140000,
        25: 160000, 26: 250000
}
df['income_cont'] = df['income16'].map(income_map)

df['binary_lvmus']  = (df['yrlvmus']  == 1).astype(int)
df['binary_artxbt'] = (df['yrartxbt'] == 1).astype(int)
```

```
df['binary_movie']  = (df['yrmovie']  == 1).astype(int)
df['binary_creat']  = (df['yrcreat']  == 1).astype(int)
df['omni'] = df[['binary_lvmus','binary_artxbt','binary_movie','binary_creat']].
  ↪sum(axis=1)
```

# 2 Investigating the relation between age and annual income

## 2.1 Calculating covariance

[ ]:
```
# np.cov returns a 2x2 matrix; [0,1] is the off-diagonal covariance
cov_val = np.cov(df['age'], df['income_cont'])[0, 1]
print(f'Covariance(age, income): {cov_val:.2f}')
np.cov(df['age'], df['income_cont'])
```

## 2.2 Data Visualisation

[ ]:
```
(
        p9.ggplot(df, p9.aes(x='age', y='income_cont')) +
        p9.geom_point(alpha=0.3) +
        p9.geom_smooth(method='lm', se=False, color='steelblue') +
        p9.theme_light() +
        p9.labs(x='Age', y='Annual Income ($)')
)
```

## 2.3 Bivariate statistic

Pearson correlation — appropriate for two continuous variables. Comment on the magnitude, direction, and p-value.

[ ]:
```
r, p = stats.pearsonr(df['age'], df['income_cont'])
print(f'Pearson r = {r:.4f}, p-value = {p:.4f}')
```

# 3 Investigating income and cultural omnivorousness

## 3.1 Data Visualisation

Omnivorousness is ordinal (0–4), so a boxplot is appropriate.

[ ]:
```
(
        p9.ggplot(df, p9.aes(x='factor(omni)', y='income_cont')) +
        p9.geom_boxplot() +
        p9.theme_light() +
        p9.labs(x='Omnivorousness Index', y='Annual Income ($)')
)
```

## 3.2 Bivariate statistic

Spearman correlation — appropriate when one variable is ordinal.

[ ]:
```
rho, p = stats.spearmanr(df['omni'], df['income_cont'])
print(f'Spearman rho = {rho:.4f}, p-value = {p:.4f}')
```

# 4 Investigating race and highest education level attained

## 4.1 Recoding variables

[ ]:
```
df['race_recoded']   = df['race'].map({1: 'White', 2: 'Black'}).fillna('Other')
df['degree_recoded'] = df['degree'].map({
        0: '1:<HS', 1: '2:HS', 2: '3:Assoc.', 3: '4:BA', 4: '5:Grad.'
})
```

## 4.2 Preliminary visualisation

[ ]:
```
(
        p9.ggplot(df) +
        p9.geom_bar(p9.aes(x='degree_recoded', fill='race_recoded')) +
        p9.theme_light() +
        p9.labs(x='Degree', y='Count', fill='Race')
)
```

[ ]:
```
(
        p9.ggplot(df) +
        p9.geom_bar(p9.aes(x='degree_recoded', fill='degree_recoded')) +
        p9.facet_wrap('race_recoded') +
        p9.theme_light() +
        p9.labs(x='Degree', y='Count', fill='Degree')
)
```

## 4.3 Bivariate statistic: Chi-square and Cramér's V

Chi-square tests if there is a statistically significant association. Cramér's V measures the strength of that association (0 = no association, 1 = perfect).

[ ]:
```
contingency = pd.crosstab(df['race_recoded'], df['degree_recoded'])

chi2_result = stats.chi2_contingency(contingency)
cramerv_result = stats.contingency.association(contingency, method='cramer')
```